Homework 06: Sample solution

QMIR

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```
start_time <- Sys.time()

library(tidyverse) # wrangling and visualization

set.seed(3594)
```

```r
# Shared course palette.
col_1 <- "steelblue"
col_2 <- "#E07B39"
col_3 <- "#7B3F9E"
col_4 <- "goldenrod3"

# The probe log: D is a democracy, N is not.
probe_log <- str_split_1("DNNDNNDDNNNDNDNNDNND", "")

# Grid posterior: Beta(a, b) prior, zero below `lower`.
posterior_grid <- function(W, N, a = 1, b = 1, lower = 0) {
  tibble(theta = ((0:999) + 0.5) / 1000) |>
    mutate(
      prior = dbeta(theta, a, b) * (theta >= lower),
      prior = prior / sum(prior),
      likelihood = dbinom(W, size = N, prob = theta),
      posterior = prior * likelihood / sum(prior * likelihood)
    )
}

# Prior, likelihood and posterior in one plot.
plot_update <- function(W, N, a = 1, b = 1, lower = 0) {
  posterior_grid(W, N, a, b, lower) |>
    mutate(likelihood = likelihood / sum(likelihood)) |>
    pivot_longer(
      c(prior, likelihood, posterior),
      names_to = "curve",
      values_to = "probability"
    ) |>
    mutate(
      curve = factor(curve, levels = c("prior", "likelihood", "posterior")),
      density = probability * 1000
    ) |>
    ggplot(aes(x = theta, y = density, colour = curve, linetype = curve)) +
    geom_line(linewidth = 0.8) +
    scale_colour_manual(values = c(col_1, col_4, col_2)) +
    scale_linetype_manual(values = c("dashed", "dotted", "solid")) +
    theme_classic() +
    theme(legend.position = "bottom") +
    labs(
      x = expression(theta ~ "(share of land in democracies)"),
      y = "Density",
```

```
      colour = NULL,
      linetype = NULL,
      title = str_glue("Beta({a}, {b}) prior, {W} democratic landings in {N}")
    )
}
```

1 The setting

The setting introduces the aliens, the probe log and the two helpers. It asks no questions.

2 The estimand

1. Say in your own words what θ is. Is it the same as the share of the world's **countries** that are democracies? Is it the same as the share of the world's **people** who live in a democracy? Use Russia and India to explain why the three numbers differ.

θ is the share of Earth's land surface that lies in countries V-Dem classifies as electoral or liberal democracies. It weighs every country by its **area**. The share of countries weighs every country equally, and the share of people weighs every country by its **population**. Russia is one country among about 180, but it covers about a ninth of the world's land and holds under 2 percent of its people. India holds about a sixth of the world's people on about 2 percent of its land. So the classification of Russia moves θ a lot, the classification of India hardly at all, and for the share of people it is the other way round, while for the share of countries both count exactly once. The three numbers answer three different questions.

2. Which of the three would you want to know if your question were "how many people can vote their government out?", and which if it were "how much of the planet's territory, resources and borders do democracies control?"

The first question is about people, so it needs the share of people living in democracies. The second is about territory, so it needs θ, the share of land. The estimand follows from the question, never from whatever is easiest to measure.

3. About a tenth of Earth's land is Antarctica, which belongs to no country. What should a probe record when it lands there? Say how your choice changes what θ means.

There are three defensible choices, and each defines a different θ.

- **Record N.** Antarctica is not a democracy, so θ becomes the share of **all** land in democracies, and it can be at most about 0.9.

- **Discard the landing and send another probe.** θ becomes the share of land **that belongs to a country** in democracies.
- **Record a third letter.** The outcome is no longer binary, and the binomial model no longer fits.

The second choice best matches the question about democracies, because Antarctica has no regime to classify. What matters most is that the choice is made and written down **before** the probes land, because it is part of step 1, the estimand.

4. In the session, the parameter has one true value, but we do not know it. Apply this to θ: in what sense is it fixed, and in what sense is it uncertain? Would sending more probes change θ? What does it change instead?

θ is fixed: at a given date, with given borders and V-Dem's codings, there is one definite share of land in democracies. It is uncertain to the aliens, because they do not know that share, and a probability distribution over θ describes what they know about it. More probes do not change θ, because they do not move borders or regimes. They change the aliens' **knowledge** of θ: the posterior. θ itself changes only when the world does, for instance when a country democratises, and then it is a different estimand dated to a different year.

5. The aliens treat landings as independent, each D with the same probability θ. Name one way this could fail in practice.

Any one of these is a good answer.

- **The landing spots are not uniform.** If probes avoid ice, mountains or deserts, they undersample Greenland, Siberia or the Sahara, and the landings estimate the share of reachable land instead of θ.
- **The landings are not independent.** If probes are dropped in batches from one mothership over one hemisphere, nearby landings tend to hit the same country.
- **The outcome is misrecorded.** A probe near a border may identify the wrong country.

3 Probability warm-up

Suppose, for this section only, that $\theta = 0.4$, and a mission sends 20 probes.

1. **Before running anything:** what is the most likely number of D landings? Would you expect to see exactly that number most of the time?

The most likely count is $0.4 \times 20 = 8$. I would not expect exactly 8 most of the time, because 7, 9, 6 and 10 are almost as likely, and together they take most of the probability.

2. Compute the probability of exactly 8 D landings with `dbinom()`, and the probability of at most 5 with `pbinom()`.

```r
dbinom(8, size = 20, prob = 0.4)
```

```
[1] 0.1797058
```

```r
pbinom(5, size = 20, prob = 0.4)
```

```
[1] 0.125599
```

The probability of exactly 8 is 0.18, and the probability of at most 5 is 0.126.

3. The most likely count has a probability far below one half. Explain how both can be true.

The probability is spread over 21 possible counts, from 0 to 20. 8 gets more than any other single count, but the neighbouring counts get almost as much, so 8 holds only about a fifth of the total. "Most likely" compares counts with each other, while "likely" compares one count with all the others together.

4. Now approximate the probability of at most 5 by simulating 10,000 missions of 20 probes with `rbinom()` and counting. Why do the simulated and the computed answer agree, and why not exactly? What would happen with a million missions?

```r
mean(rbinom(1e4, size = 20, prob = 0.4) <= 5)
```

```
[1] 0.1266
```

Both answer the same question. `pbinom()` computes the probability exactly, and the simulation counts how often it happens in many repetitions, which approximates it as a long-run frequency. They differ slightly because 10,000 missions are a random sample of missions, and another seed gives a slightly different share. With a million missions the share would land closer to 0.126, as in the long-run figure of the session, although each extra mission helps less than the one before.

5. `dbeta(0.5, 20, 20)` returns a number above 5, while no value of `dbinom()` can ever exceed 1. Run it, then explain why the first is not a mistake.

```r
dbeta(0.5, 20, 20)
```

```
[1] 5.014828
```

`dbinom()` gives the probability of a count, which is discrete, so every value is a probability and lies between 0 and 1. `dbeta()` gives a **density**, the height of a curve over a continuous quantity. Only areas under a density are probabilities. Beta(20, 20) packs almost all its mass into a narrow band around 0.5, so the curve must be tall there for its total area to be 1.

6. Every function in this section takes θ as known and tells you about the data. Write the quantity they compute in the notation of the session, $P(\cdot \mid \cdot)$. The aliens want the opposite: what they know is the data. Write down the quantity they want.

The functions compute $P(\text{data} \mid \theta)$. The aliens want $P(\theta \mid \text{data})$. Bayes' theorem is the bridge from the first to the second, and it needs a prior to cross it.

4 The four ingredients

No code and no numbers in this section.

1. For each of the four ingredients of Bayes' theorem (the **prior**, the **likelihood**, the **normalising constant** and the **posterior**), write two or three sentences:

   - what it is, in your own words
   - what job it does in Bayes' theorem
   - what it means for the aliens and their probes

- **Prior.** How plausible each value of θ is before the data. It brings in what is known, or assumed, before looking. For the aliens it is their belief about the share of democratic land before any probe lands, for example from intercepted broadcasts.
- **Likelihood.** How probable the observed data are under each value of θ. It scores every value by how well it accounts for what was seen. For the aliens it says, for every candidate share, how probable a log with 8 D landings in 20 would be.
- **Normalising constant.** Prior times likelihood, added up over every value of θ. It is the probability of the data averaged over everything the aliens thought possible, and dividing by it makes the posterior add up to 1. For the aliens it is the overall probability of seeing exactly this probe log under their model and prior.
- **Posterior.** How plausible each value of θ is after the data. It combines prior and likelihood, and it is the answer of the analysis. For the aliens it is what they now believe about the share of democratic land, and how sure they are.

2. Write Bayes' theorem for the aliens, with θ and "the probe log", and label each of its four terms.

$$\underbrace{P(\theta \mid \text{probe log})}_{\text{posterior}} = \frac{\overbrace{P(\text{probe log} \mid \theta)}^{\text{likelihood}}\ \overbrace{P(\theta)}^{\text{prior}}}{\underbrace{P(\text{probe log})}_{\text{normalising constant}}} \tag{1}$$

3. The likelihood gives a probability for every value of θ, yet these probabilities need not add up to 1. Why not? What do they add up to 1 over instead?

Each value of the likelihood comes from a different distribution: the distribution of the data for one particular θ. Adding them up over θ mixes probabilities from different distributions, so nothing forces the sum to be 1. For a fixed θ, the probabilities add up to 1 over all possible **data**, here all counts from 0 to 20.

4. The normalising constant is the same number for every value of θ. Why? And if it is the same everywhere, why do we need it at all?

It is a sum over every value of θ, so θ has been summed out and the result cannot depend on it. Dividing by it does not change which values are more plausible than which, or the shape of the posterior. It turns the scores into probabilities that add up to 1, and only then can the aliens make statements such as "the probability that θ is above 0.5 is 0.19".

5 Counting the ways

The aliens start with a crude model: Earth as a **four-faced die**, each face democratic or not. There are five conjectures, $\theta \in \{0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, 1\}$, where k is the number of democratic faces. The first three probes of the log landed **D N N**.

Do this section by hand, as in the session. Fill in the table directly in this document.

1. With k democratic faces, in how many ways can D N N happen? Write the general expression, then fill in the table with every conjecture equally plausible beforehand.

The first D can land on any of k democratic faces, and each N on any of $4-k$ other faces, so D N N can happen in $k \times (4-k) \times (4-k) = k(4-k)^2$ ways.

Table 1: The four-faced alien globe after D N N, from a flat prior. Each row of prior times ways is divided by their sum, 20.

k	θ	Prior	Ways to see D N N	Prior times ways	Posterior
0	0	1	0	0	0.00
1	1/4	1	9	9	0.45
2	2/4	1	8	8	0.40

k	θ	Prior	Ways to see D N N	Prior times ways	Posterior
3	3/4	1	3	3	0.15
4	1	1	0	0	0.00
				Sum: 20	**Sum: 1**

2. Which column of Table 1 is which of the four ingredients? Where is the normalising constant?

The prior column is the **prior**, the ways column is the **likelihood** (up to a constant, see question 7), and the last column is the **posterior**. The normalising constant is not a column: it is the **sum** of prior times ways, 20, which every row is divided by.

3. Which conjectures are ruled out, and by which landing? Why can no later landing bring them back? What is this called?

The first D rules out $k = 0$, because an all-N globe has no way to produce a D. The first N rules out $k = 4$, for the same reason. Every later landing multiplies its ways onto that zero, so those two rows stay zero for good. This is **falsification**: a zero likelihood, where the data are impossible under a conjecture.

4. The fourth landing is **D**. Update by multiplying your counts from Table 1 by the ways a single D can happen under each conjecture. Then count the ways for all four landings, D N N D, at once. Show that both routes give the same posterior, and explain why the order of updating cannot matter.

A single D can happen in k ways.

Table 2: Updating the four-faced globe with a fourth landing, D. Multiplying the old counts by the ways for the new D gives the same counts as counting all four landings at once.

k	Ways for D N N	Ways for the new D	Updated ways	All at once, $k^2(4-k)^2$	Posterior
0	0	0	0	0	0.000
1	9	1	9	9	0.265
2	8	2	16	16	0.471
3	3	3	9	9	0.265
4	0	4	0	0	0.000
			Sum: 34	**Sum: 34**	**Sum: 1**

Both routes give the counts 0, 9, 16, 9, 0 in Table 2, so both give the same posterior. The order cannot matter because the counts are a product, and a product is the same in any order.

Yesterday's posterior is today's prior, and learning one probe at a time ends where learning from all probes at once does.

5. Start again from D N N, now with the prior 1, 2, 3, 2, 1 for $k = 0, \ldots, 4$. What belief does this prior encode? Recompute the posterior and say in one or two sentences what the prior changed and what it could not change.

The prior says that a half-democratic globe is the most plausible, that lopsided globes are less plausible, and that neither side is favoured.

Table 3: The four-faced alien globe after D N N, from the prior 1, 2, 3, 2, 1.

k	Prior	Ways to see D N N	Prior times ways	Posterior
0	1	0	0	0.000
1	2	9	18	0.375
2	3	8	24	0.500
3	2	3	6	0.125
4	1	0	0	0.000
			Sum: 48	**Sum: 1**

The prior moved the most plausible conjecture from $k = 1$ to $k = 2$ (Table 3), because three landings are too few to outweigh it. It could not bring back $k = 0$ or $k = 4$: the data ruled them out, and no prior weight can multiply a zero likelihood into something positive.

6. `posterior_grid()` in the setup chunk does on 1000 conjectures what your table did on five. Point to the line that computes each of the four ingredients.

- **Prior:** `prior = dbeta(theta, a, b) * (theta >= lower)`, then `prior = prior / sum(prior)`.
- **Likelihood:** `likelihood = dbinom(W, size = N, prob = theta)`.
- **Normalising constant:** `sum(prior * likelihood)`, the denominator in the next line.
- **Posterior:** `posterior = prior * likelihood / sum(prior * likelihood)`.

7. **Optional check in R:** `dbinom(1, size = 3, prob = (0:4) / 4)` is proportional to your ways column. Why proportional and not equal?

```
dbinom(1, size = 3, prob = (0:4) / 4)
```

```
[1] 0.000000 0.421875 0.375000 0.140625 0.000000
```

```
dbinom(1, size = 3, prob = (0:4) / 4) * 4^3 / 3
```

```
[1] 0 9 8 3 0
```

The binomial probability counts all three orders that give one D in three (D N N, N D N, N N D) and divides by all $4^3 = 64$ paths three landings can take. So it equals the ways times $3/64$, the same factor for every k. The factor cancels when the posterior is normalised, which is why ways and binomial probabilities give the same posterior.

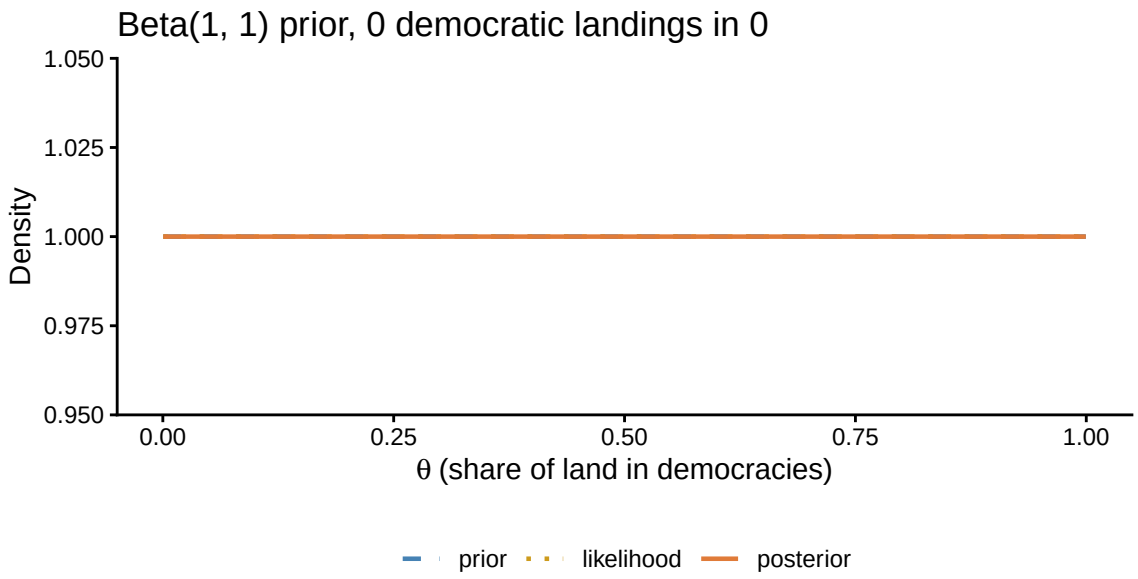
6 Priors as commitments

Three alien scientists disagree before any probe lands. Each writes down a Beta prior for θ, the same priors as on the dashboard:

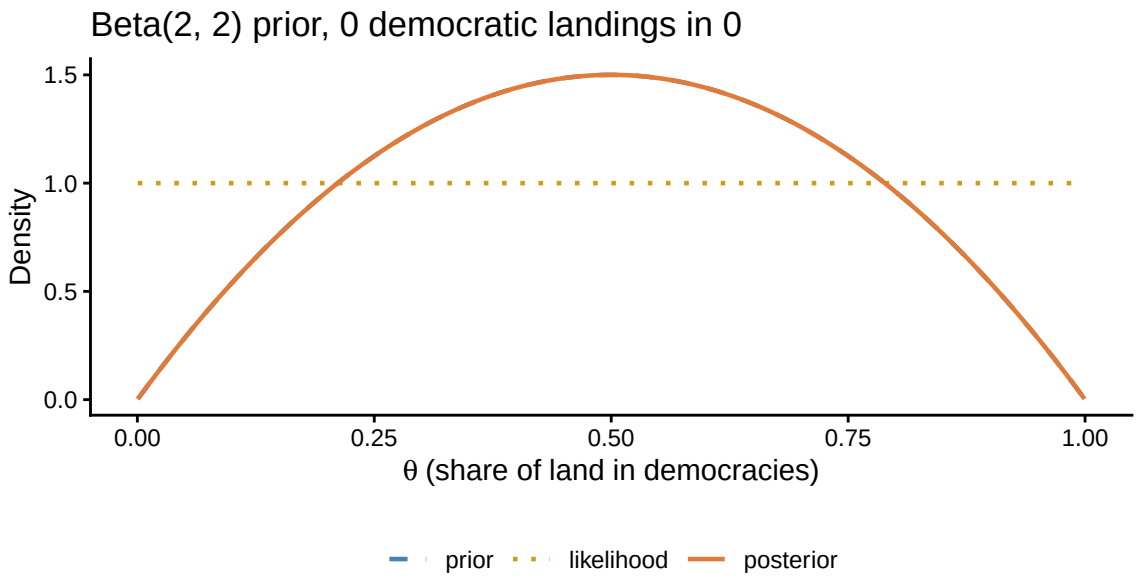
- **Uniform**, Beta(1, 1)
- **Weakly informative**, Beta(2, 2)
- **Dogmatic**, Beta(500, 500)

1. Draw each prior alone with `plot_update(W = 0, N = 0, a = , b = )`. Why does the posterior equal the prior when `N = 0`?

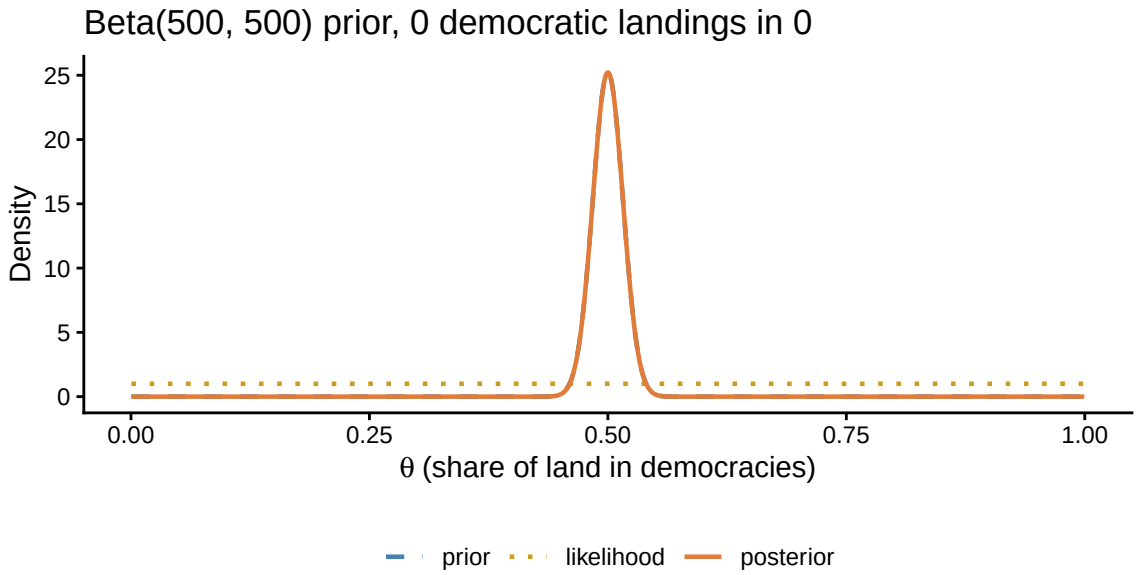
```
plot_update(W = 0, N = 0, a = 1, b = 1)
```




```
plot_update(W = 0, N = 0, a = 2, b = 2)
```




```
plot_update(W = 0, N = 0, a = 500, b = 500)
```



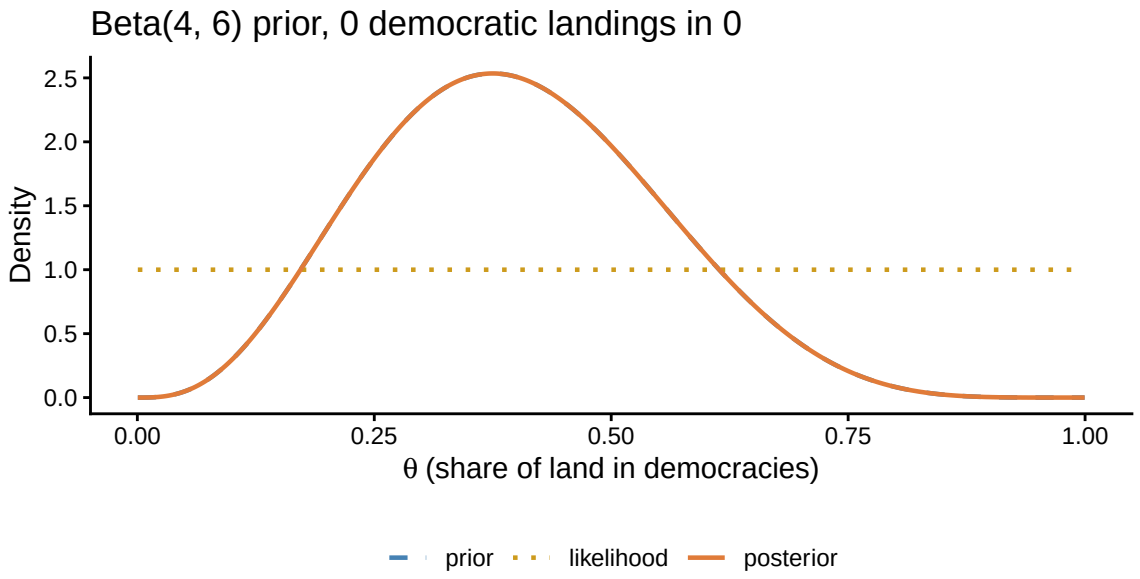
With no landings, the probability of the (empty) data is 1 under every value of θ, so the likelihood is flat. Multiplying the prior by a constant and normalising gives the prior back. Without data there is nothing to learn from.

2. For each prior, one or two sentences: what does the scientist commit to about θ, and could the scientist defend it to a political scientist on Earth?

- **Uniform.** Every share from 0 to 1 is equally plausible, including a world that is almost all democratic or almost all autocratic. It is defensible as a statement of complete ignorance, but an Earth political scientist knows that both regime types hold large territories, so it throws information away.
- **Weakly informative.** Shares around one half are the most plausible, and shares near 0 or 1 are unlikely but not impossible (90 percent of its mass lies between 0.14 and 0.86). It is easy to defend: it encodes only that both regime types hold a sizeable part of the land.
- **Dogmatic.** The share is almost certainly between 0.474 and 0.526, as sure as if the scientist had already seen about 1000 probes split evenly. It cannot be defended without such evidence, and intercepted broadcasts say nothing about land area.

3. Now choose your own prior for θ. Pick `a` and `b`, draw it, and defend it in three to four sentences on principled grounds: what do you know about Earth before any probe lands, and how sure are you? The Beta mean is $a/(a+b)$, and a larger $a+b$ makes it more concentrated.

```
plot_update(W = 0, N = 0, a = 4, b = 6)
```



Many priors are defensible, and the defence is what counts. One example is Beta(4, 6), with mean 0.4 and 90 percent of its mass between 0.17 and 0.66. The two largest countries, Russia and China, are autocracies, and so is much of Central Asia, the Middle East and Africa, while Canada, the United States, Brazil and Australia are large democracies. That suggests somewhat

less than half, so the mean sits below 0.5. I am not sure of it, so $a + b = 10$ makes the prior worth only about ten landings, and a few dozen probes can overrule it.

7 Prior versus data

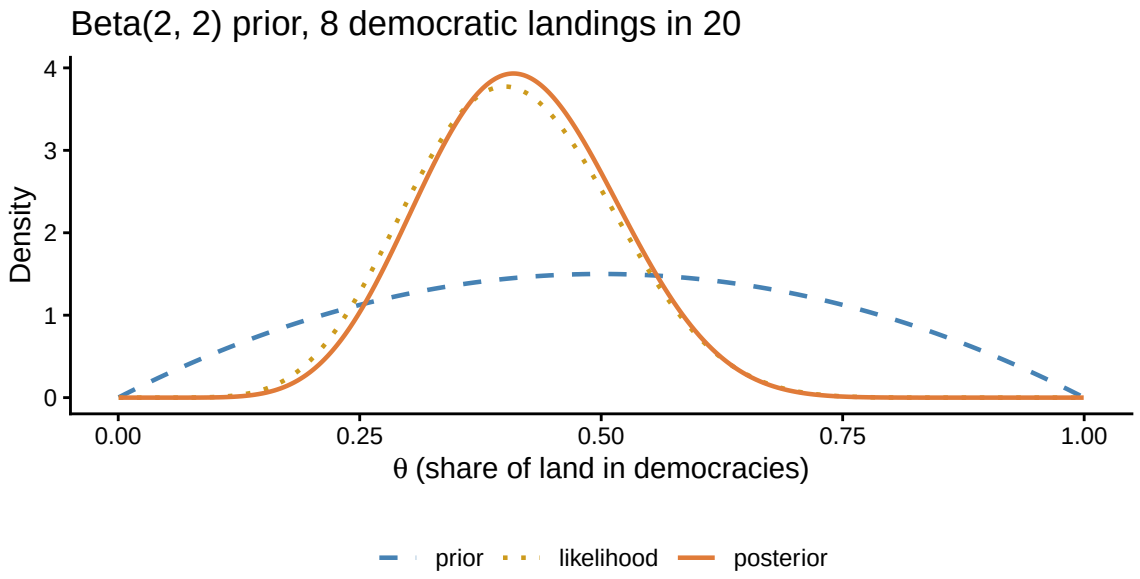
The log has 8 D landings in 20. Suppose the aliens send more probes and keep seeing the same share: 80 in 200, then 800 in 2000.

1. **Predict first:** with the weakly informative and the dogmatic prior, where will the posterior sit after 20, 200 and 2000 landings, and how wide will it be? Write your prediction down before you run anything.

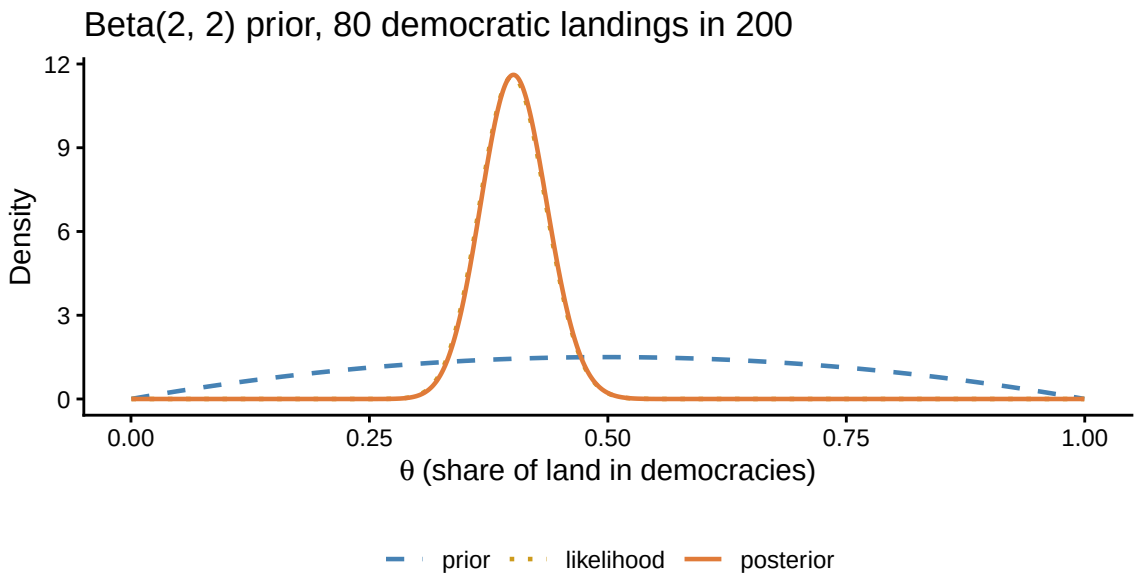
A good prediction: under the weakly informative prior, the posterior sits near 0.4 from the start and gets narrower with every tenfold increase in data. Under the dogmatic prior, it starts at 0.5, hardly moves at 20 landings, and drifts towards 0.4 only with thousands of landings.

2. Run `plot_update()` for both priors and all three sample sizes.

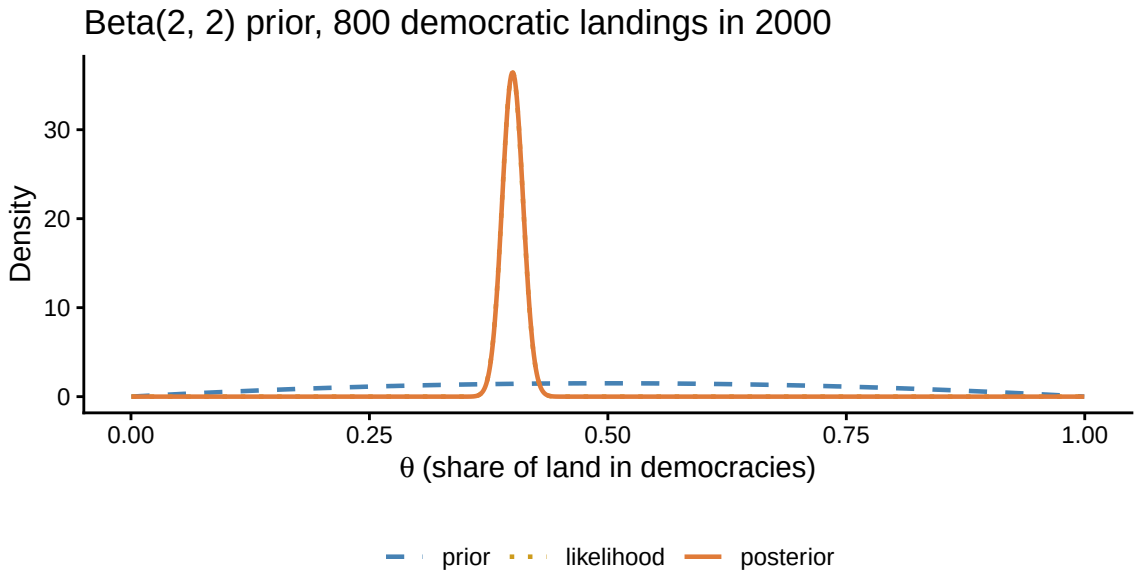
```
plot_update(W = 8, N = 20, a = 2, b = 2)
```



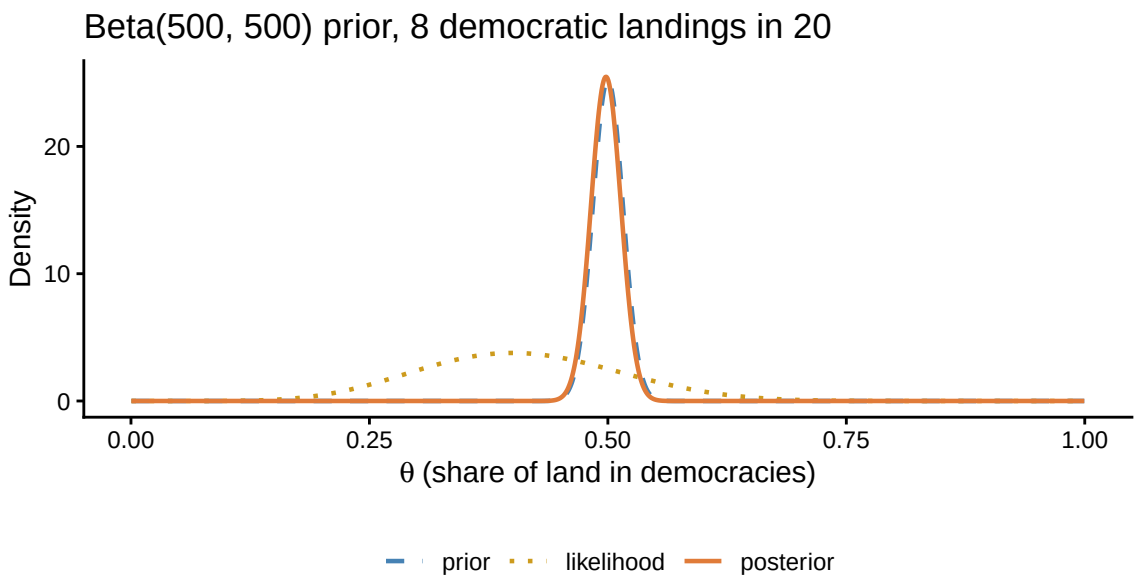

```
plot_update(W = 80, N = 200, a = 2, b = 2)
```



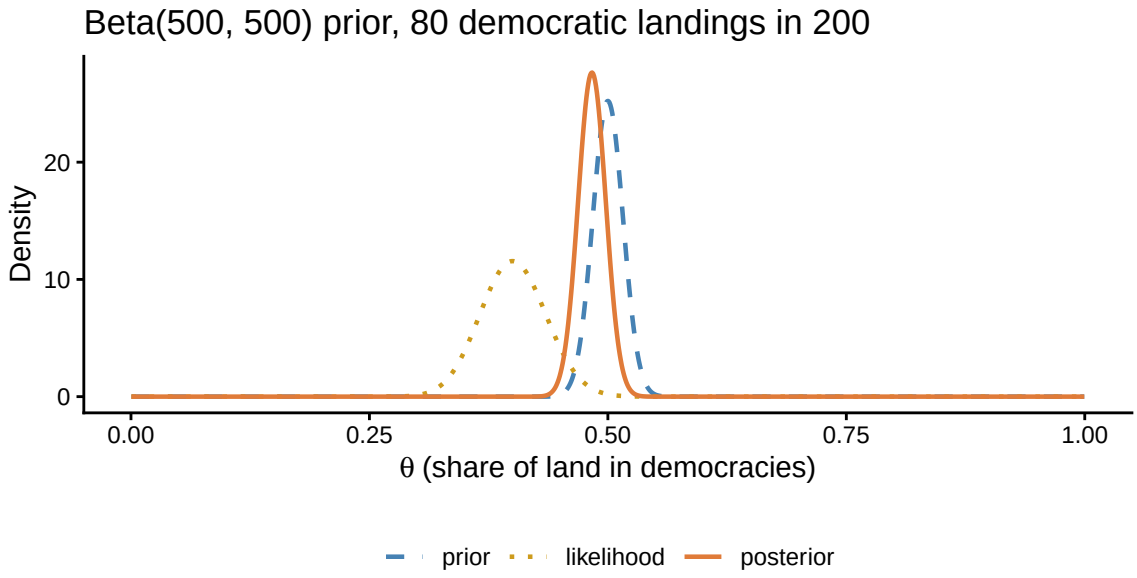

```
plot_update(W = 800, N = 2000, a = 2, b = 2)
```



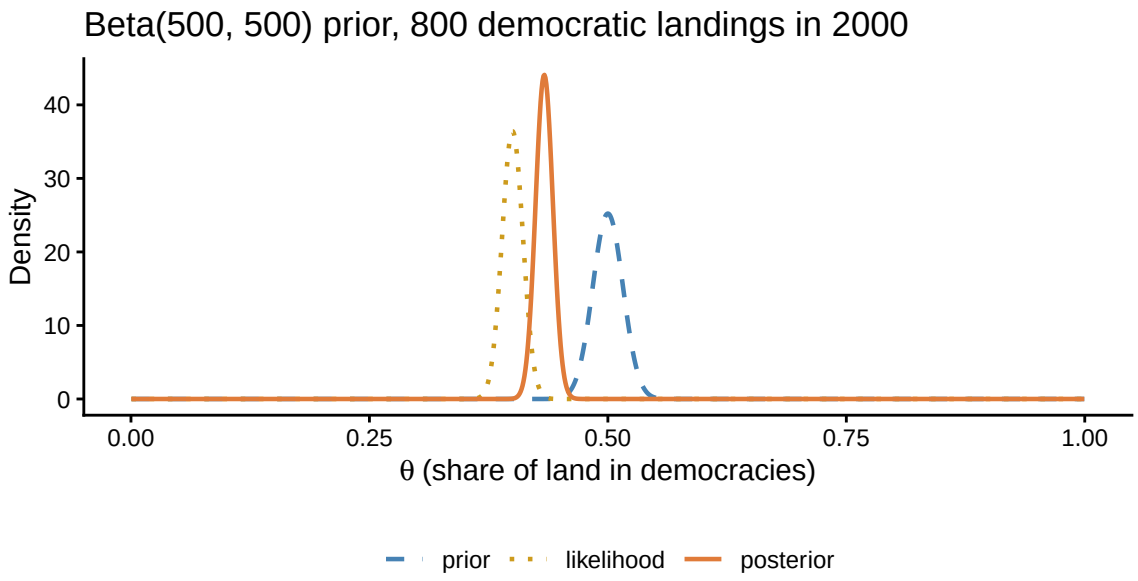

```
plot_update(W = 8, N = 20, a = 500, b = 500)
```




```
plot_update(W = 80, N = 200, a = 500, b = 500)
```




```
plot_update(W = 800, N = 2000, a = 500, b = 500)
```




```
expand_grid(
  tibble(prior = c("Beta(2, 2)", "Beta(500, 500)"), a = c(2, 500)),
  N = c(20, 200, 2000)
) |>
  mutate(summary = map2(a, N, \(a, N) grid_summary(round(0.4 * N), N, a, a))) |>
  unnest(summary) |>
  select(prior, landings = N, mean, sd) |>
  knitr::kable(digits = 3)
```

Table 4: Posterior mean and standard deviation of theta for two priors, when 40 percent of the landings are democratic. Under the weakly informative prior the mean is near 0.4 from the start, under the dogmatic prior it moves only slowly.

prior	landings	mean	sd
Beta(2, 2)	20	0.417	0.099
Beta(2, 2)	200	0.402	0.034
Beta(2, 2)	2000	0.400	0.011
Beta(500, 500)	20	0.498	0.016
Beta(500, 500)	200	0.483	0.014
Beta(500, 500)	2000	0.433	0.009

3. Where was your prediction right, and where wrong?

This depends on the prediction. A common surprise is how little the dogmatic prior moves:

after 2000 landings its posterior mean is still 0.43 (Table 4), well away from the observed 0.4. Another is that its posterior is narrow from the start, so it looks certain while being wrong.

4. Explain **why** the posterior moves the way it does as the data grow. Use the four ingredients, not a description of the curves.

The posterior is prior times likelihood, normalised. The prior stays the same, but every landing multiplies another factor into the likelihood, so the likelihood gets narrower and more peaked around the observed share with every landing. With few landings the prior and the likelihood are comparable and the posterior is a compromise. With many, the likelihood is so concentrated that it outweighs the prior wherever the prior is not tiny, so the posterior sits where the data say and is about as narrow as the likelihood.

5. Under the weakly informative prior, the prior barely matters at 2000 landings. Under the dogmatic prior, it still matters. What is it about a prior that decides how much data it takes to overrule it?

Its **concentration**, together with how far it sits from what the data say. For a Beta prior, $a + b$ acts roughly like a number of landings already seen: Beta(2, 2) is worth about 4 landings, Beta(500, 500) about 1000. Data overrule a prior once they are worth clearly more than it, so the dogmatic prior needs many thousands of landings to give way, and the further its centre is from the truth, the more it takes.

8 Zero never learns

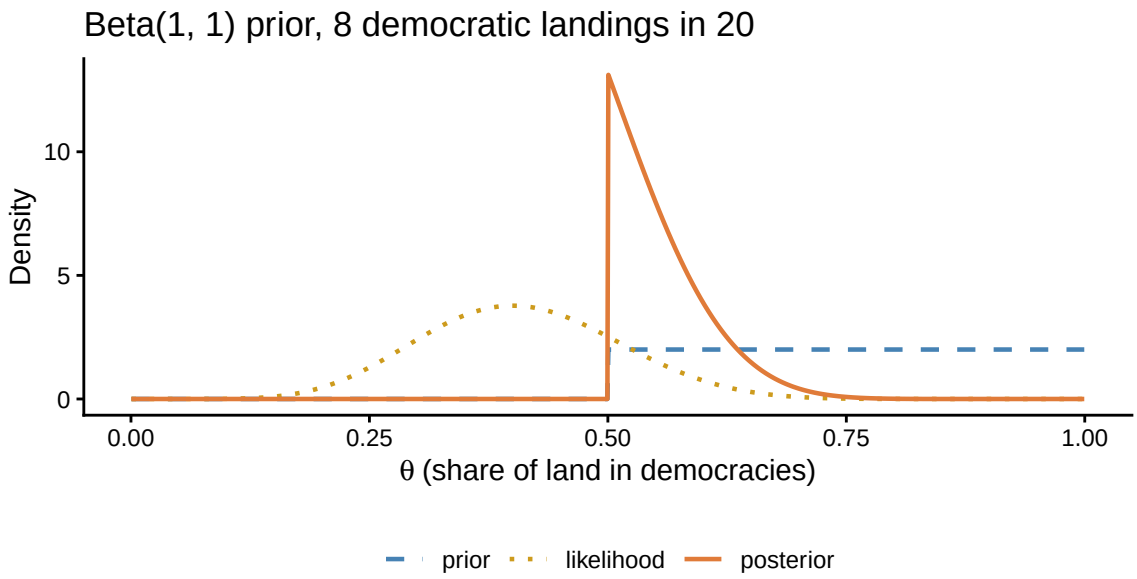
A senior scientist is certain that democracies hold at least half of Earth's land. Her prior is uniform on $[0.5, 1]$ and exactly zero below 0.5. `plot_update(W, N, lower = 0.5)` draws it.

1. **Predict first:** where will her posterior sit after 8 D landings in 20, and after 800 in 2000?

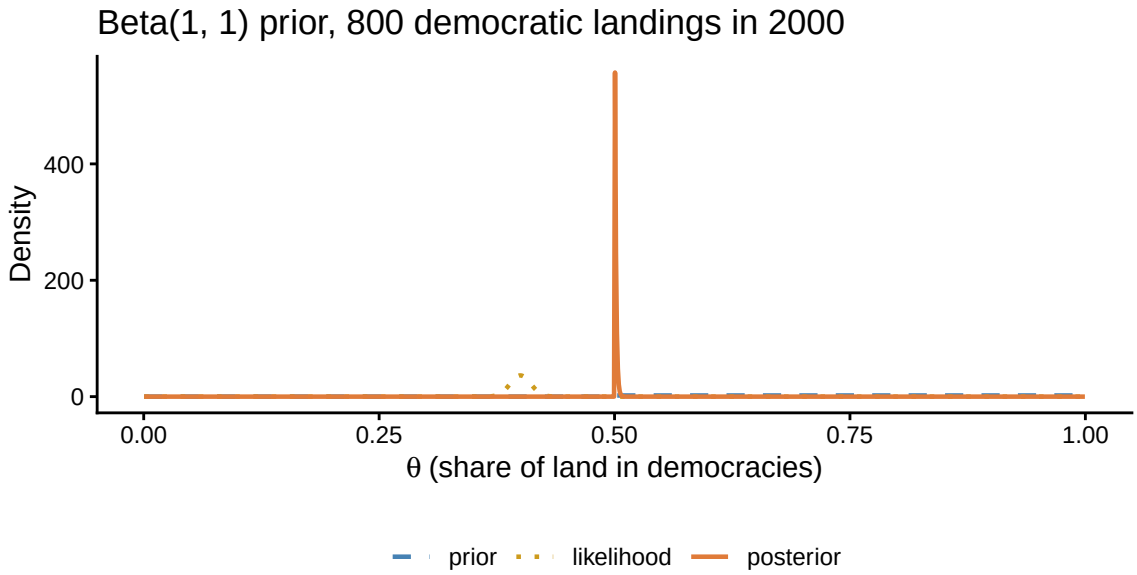
A good prediction: it cannot go below 0.5, so it piles up just above 0.5, and more tightly with more data.

2. Run both.

```
plot_update(W = 8, N = 20, lower = 0.5)
```




```
plot_update(W = 800, N = 2000, lower = 0.5)
```



After 20 landings her posterior mean is 0.559. After 2000 it is 0.501, pressed against the boundary.

3. Explain what you see with Bayes' theorem. Why does her posterior pile up against 0.5, however many probes land?

The posterior is the likelihood times the prior, normalised. Below 0.5 the prior is zero, so the posterior is zero there whatever the likelihood says. Above 0.5 the likelihood is highest where θ is closest to the observed 0.4, which is right at 0.5. With more data the likelihood falls off ever more steeply away from 0.4, so the posterior is squeezed ever closer against 0.5. The data keep saying "lower", and the prior allows nothing lower.

4. In Section 5, a likelihood of zero ruled out two conjectures. Here a prior of zero rules out half of them. Both end with a posterior of zero. Explain the difference between the two, and say which is falsification and which is dogma.

A zero likelihood means the data observed are impossible under a conjecture: an all-N globe cannot produce a D. The conjecture is ruled out **by evidence**, after looking, and that is falsification. A zero prior rules a conjecture out **before** looking, and no evidence can bring it back. That is dogma.

5. When is a prior of exactly zero defensible? Give one statement about θ that deserves a prior of exactly zero, and explain why.

Only for what is logically impossible. "θ is below 0 or above 1" deserves a prior of zero, because a share of land cannot be negative or larger than the whole. A Beta prior builds this in, because it lives on $[0, 1]$.

6. Rewrite her belief so that it respects Cromwell's rule but still says that more than half is likely. Give `a` and `b` and say why.

Beta(14, 6), the informative prior from the dashboard, has mean 0.7 and puts 0.97 of its mass above 0.5. It says "more than half is very likely", but it gives every share between 0 and 1 some plausibility. With the 20 landings of the log, her posterior probability for more than half falls to 0.74, and with 800 in 2000 it falls to practically zero: she can now be persuaded.

9 Answering the question

The aliens' question was: **do democracies hold more than half of Earth's land?**

1. Compute $P(\theta > 0.5 \mid \text{probe log})$ for the 20 landings of the log, once with the uniform prior and once with your own prior from Section 6. One short pipeline each is enough. Hint: filter the rows of `posterior_grid()` with `theta > 0.5` and add up their `posterior`.

```
posterior_grid(8, 20) |>
  filter(theta > 0.5) |>
  summarise(sum(posterior))
```

```
# A tibble: 1 x 1
  `sum(posterior)`
             <dbl>
1            0.192
```

```
posterior_grid(8, 20, a = 4, b = 6) |>
  filter(theta > 0.5) |>
  summarise(sum(posterior))
```

```
# A tibble: 1 x 1
  `sum(posterior)`
             <dbl>
1            0.132
```

With the uniform prior, $P(\theta > 0.5 \mid \text{probe log}) \approx 0.19$. With Beta(4, 6) it is about 0.13.

2. Answer the aliens' question in one or two sentences, in plain language, with your own prior.

Probably not: given the 20 landings, the binomial model and our Beta(4, 6) prior, the probability that democracies hold more than half of Earth's land is about 0.13. That is roughly one chance in 8, so more probes would be needed to be confident either way.

3. Three junior scientists summarise the result with the uniform prior. For each, say whether it is right, and if not, what exactly is wrong.

    - **(a)** "Given our 20 landings, our binomial model and a uniform prior, the probability that democracies hold more than half of Earth's land is about 0.19."
    - **(b)** "If democracies held more than half of Earth's land, there would be a 0.19 probability of seeing a probe log like ours."
    - **(c)** "The probability is below one half, so democracies hold less than half of Earth's land."

- **(a) is right.** It is a graded statement about the hypothesis itself, $P(\theta > 0.5 \mid \text{data})$, and it names all three things it depends on: the data, the model and the prior.
- **(b) is wrong.** It swaps the conditional: it describes $P(\text{data} \mid \theta > 0.5)$, a probability of the data given the hypothesis. The 0.19 is the probability of the hypothesis given the data, and the two are different numbers.
- **(c) is wrong.** It turns a graded statement into a verdict. A probability of 0.19 means that more than half is unlikely, not that it is ruled out: about one chance in five remains. It also drops what the probability is conditional on.

10 Why grids explode

1. The aliens now want θ separately for each of six continents, with a grid of 1000 values for each. How many terms does the normalising constant add up? Write the number as a power of 10.

Every combination of the six shares needs its own term, so there are $1000^6 = (10^3)^6 = 10^{18}$ terms.

2. In Section 5 you could count every path by hand. Why does that stop working here, and what does the session say the aliens should do instead?

On the four-faced globe there were five conjectures, and here there are 10^{18}. Even at a billion terms per second, adding them up would take about 30 years, and every extra parameter multiplies that by 1000. The session's answer, and next week's topic, is to **draw samples** from the posterior instead of computing it everywhere, with Markov chain Monte Carlo. The normalising constant is then never computed, and counting samples replaces counting ways.

Reflection

Which idea from this homework was the hardest to put in your own words? Is there a part of Bayes' theorem you are still unsure about?

The reflection is personal and has no sample answer. A good one names a concrete idea, such as why the likelihood need not add up to 1, and says what made it hard.

Session info and execution time

```
sessionInfo()
```

```
R version 4.6.1 (2026-06-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
```

```
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: Europe/Berlin
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1
 [5] purrr_1.2.2     readr_2.2.0     tidyr_1.3.2     tibble_3.3.1
 [9] ggplot2_4.0.3   tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       jsonlite_2.0.0     compiler_4.6.1     tidyselect_1.2.1
 [5] scales_1.4.0       yaml_2.3.12        fastmap_1.2.0      R6_2.6.1
 [9] labeling_0.4.3     generics_0.1.4     knitr_1.52         pillar_1.11.1
[13] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.3.0        stringi_1.8.9
[17] xfun_0.60          S7_0.2.2           otel_0.2.0         timechange_0.4.0
[21] cli_3.6.6          withr_3.0.3        magrittr_2.0.5     digest_0.6.39
[25] grid_4.6.1         hms_1.1.4          lifecycle_1.0.5    vctrs_0.7.3
[29] evaluate_1.0.5     glue_1.8.1         farver_2.1.2       rmarkdown_2.32
[33] tools_4.6.1        pkgconfig_2.0.3    htmltools_0.5.9
```

```r
end_time <- Sys.time()
cat(
  "Total execution time:",
  round(difftime(end_time, start_time, units = "mins"), 1),
  "minutes\n"
)
```

```
Total execution time: 0.1 minutes
```